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Biomechanical effects of different suture pretension levels in suture-bridge repair of supraspinatus tendon tears: a finite element analysis

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Objective: The optimal initial suture pretension during suture-bridge repair of full-thickness supraspinatus tendon tears remains uncertain and is largely determined by the surgeon's experience. This study aimed to evaluate the effects of different suture pretensions on the mechanical environment of the tendon-bone interface and to provide a biomechanical reference for intraoperative tension control.

Methods: A three-dimensional finite element model of the shoulder joint was reconstructed from computed tomography data of a healthy adult volunteer. A full-thickness supraspinatus tear was simulated and repaired using a double-row suture-bridge configuration. Seven total pretension levels (30–90 N, in 10 N increments) and three shoulder abduction positions (10°, 20°, and 30°) were combined to generate 21 loading conditions. The mean contact pressure, effective contact area, relative interfacial displacement, and tendon-side von Mises stress were analyzed.

Results: Increasing pretension increased the mean tendon-bone contact pressure at all abduction angles. At 20° of abduction, the mean contact pressure increased from 0.382 MPa to 0.856 MPa. The peak tendon-side von Mises stress also increased with pretension, reaching a maximum of 4.533 MPa in the 30°/90 N condition. The effective contact area changed only slightly at 10° of abduction, decreased from 181.4 mm^2 to 169.5 mm^2 at 20°, and reached relatively high values at 50–60 N at 30°. High-stress regions were mainly concentrated around the medial-row suture perforation sites and expanded as pretension increased.

Conclusion: Suture pretension improved tendon-bone compression and initial interfacial stability, but higher pretension increased local pressure and tendon stress concentration. Within the tested configurations, 50–60 N represented an intermediate, condition-dependent range rather than a universal clinical target.

KEYWORDS
finite element analysis, pretension, rotator cuff repair, supraspinatus tendon tear, suture-bridge repair

1 Introduction

Rotator cuff tears are a common cause of shoulder pain and dysfunction, among which full-thickness tears of the supraspinatus tendon are particularly frequent. With advances in sports medicine and arthroscopic surgery, the double-row suture-bridge (SB) technique has become an important procedure for the treatment of full-thickness rotator cuff tears because it can increase the tendon-bone contact area, improve initial fixation strength, and restore footprint coverage (He et al., 2020; Tuoheti et al., 2005; Meier and Meier, 2006; Ma et al., 2006). Biomechanically, this technique uses medial- and lateral-row anchors together with bridging sutures to create a compressive construct across the tendon surface, reattaching the torn tendon to the greater tuberosity footprint and thereby improving initial tendon-bone stability and the mechanical environment required for healing (Baums et al., 2010).

Although suture-bridge repair has clear biomechanical advantages, the initial pretension applied to the sutures during surgery is still largely controlled by the surgeon's experience, and no quantitative reference standard is available. Insufficient pretension may lead to inadequate compression at the tendon-bone interface, allowing early postoperative micromotion or gap formation and compromising tendon-bone healing. Conversely, excessive pretension may concentrate stress at the suture-tendon contact region, increase tissue compression and cut-through risk, and contribute to type II retear (Golant, 2026; Lo and Burkhart, 2003; Miyake et al., 2022). Therefore, determining an appropriate level of suture pretension that balances interfacial stability and tendon tissue safety is essential for further optimization of suture-bridge repair.

A growing body of research has examined the role of suture tension in rotator cuff repair. Increased suture tension can increase contact pressure at the tendon-bone interface, whereas excessive tension may compromise tissue safety (Park et al., 2015). Zhong et al. (Zhong et al., 2025) used finite element analysis to investigate the effects of suture configuration and knot number in patch-bridging reconstruction for massive rotator cuff tears. Ntalos et al. (Ntalos et al., 2019) compared the biomechanical behavior of all-suture anchors and conventional anchors in rotator cuff repair and further evaluated the influence of insertion angle. However, most previous studies have focused on repair constructs, anchor parameters, or comparisons among double-row, transosseous, and suture-bridge techniques (Ostrander et al., 2012; Kim et al., 2016; Wang et al., 2013). Systematic analyses of tendon-bone contact behavior, local tendon stress, and interfacial micromotion under graded pretension levels remain limited, particularly under different postoperative shoulder positions (Miyake et al., 2022; Park et al., 2015; Andres et al., 2010). Moreover, few studies have simultaneously evaluated tendon-bone compression, effective contact area, interfacial micromotion, and tendon-side stress across multiple pretension levels within the same repair model.

To address this gap, the present study established a three-dimensional finite element model of a full-thickness supraspinatus tendon tear repaired using the suture-bridge technique. Seven pretension levels were evaluated at three low-abduction positions, allowing tendon-bone contact pressure, effective contact area, relative interfacial displacement, and tendon-side stress to be assessed within a consistent computational framework.

2 Materials and methods

2.1 Acquisition and reconstruction of the three-dimensional shoulder model

A 35-year-old male volunteer (height, 175 cm; weight, 70 kg) was enrolled. The volunteer had no history of shoulder injury or disease, including shoulder pain, dislocation, or fracture, and imaging examination showed no abnormality. The study protocol was approved by the Ethics Committee of the Fourth Clinical Medical College of Xinjiang Medical University (Approval No. 2025XE-GS415), under which the recruitment of one healthy adult volunteer and CT acquisition for 3D computational modeling were specifically authorized. Written informed consent was voluntarily obtained from the volunteer prior to scanning in accordance with the Declaration of Helsinki.

As it showed in Figures 1A–D, the CT images were imported into Mimics 21.0 (Materialise, Belgium), and the scapula and humerus were reconstructed using threshold segmentation, region growing, and manual correction. Because the rotator cuff tendon boundaries could not be reliably delineated by automatic threshold segmentation of the CT images, the supraspinatus and other rotator cuff structures were reconstructed geometrically rather than directly segmented. Their origins and insertions were identified from anatomical landmarks on the CT-derived scapular and humeral surfaces. The corresponding soft-tissue lengths and curvatures were estimated from the volunteer's CT images, and the geometries were manually created and adjusted in 3-matic to follow the subject-specific bony attachments and anatomical paths. Articular cartilage was constructed in HyperMesh by extruding the corresponding articular bone surfaces and adjusting the geometry between the humeral head and glenoid. The resulting models were subsequently smoothed and geometrically repaired before mesh generation and assembly.

2.2 Element types and material properties

To focus on the influence of pretension on the initial mechanical environment of the repair construct, the bone, cartilage, rotator cuff tendons, and anchors were simplified as homogeneous, isotropic, linearly elastic materials. Solid structures were discretized using four-node tetrahedral elements (C3D4), whereas sutures were modeled using two-node truss elements (T3D2) to represent their axial load-bearing behavior. The material parameters and element types implemented in the solved models are listed in Table 1 (Seki et al., 2008; Duprey et al., 2007; Sano et al., 2007; Kim et al., 2025; Quental et al., 2020). The material values were rechecked against the Abaqus input decks used for analysis, and a recent review of human bone mechanical properties was cited to provide broader context for the selected bone parameters (Verma et al., 2026).

2.3 Boundary conditions and loading

The three-dimensional model was imported into Abaqus finite element software (Dassault Systemes, USA) for boundary-condition, contact, and loading definitions, as shown in Figure 2. Based on previous finite element studies of the shoulder joint and

TABLE 1 Material properties and element types used in the finite element model.

Material	Young's modulus (MPa)	Poisson's ratio	Element type
Humerus	13,400	0.34	C3D4
Scapula	9,000	0.30	C3D4
Cartilage	400	0.46	C3D4
Rotator cuff tendons	1,200	0.40	C3D4
Anchor (PEEK)	3,800	0.28	C3D4
Suture	100	0.30	T3D2

TABLE 2 Mesh convergence results.

Mesh size	Elements	Mean CPRESS	Peak tendon stress	Mean displacement
1.5 mm	552,396	0.596 MPa	2.35 MPa	0.00915 mm
1.0 mm	1,609,968	0.611 MPa	2.443 MPa	0.00935 mm
0.8 mm	3,265,434	0.617 MPa	2.49 MPa	0.00942 mm

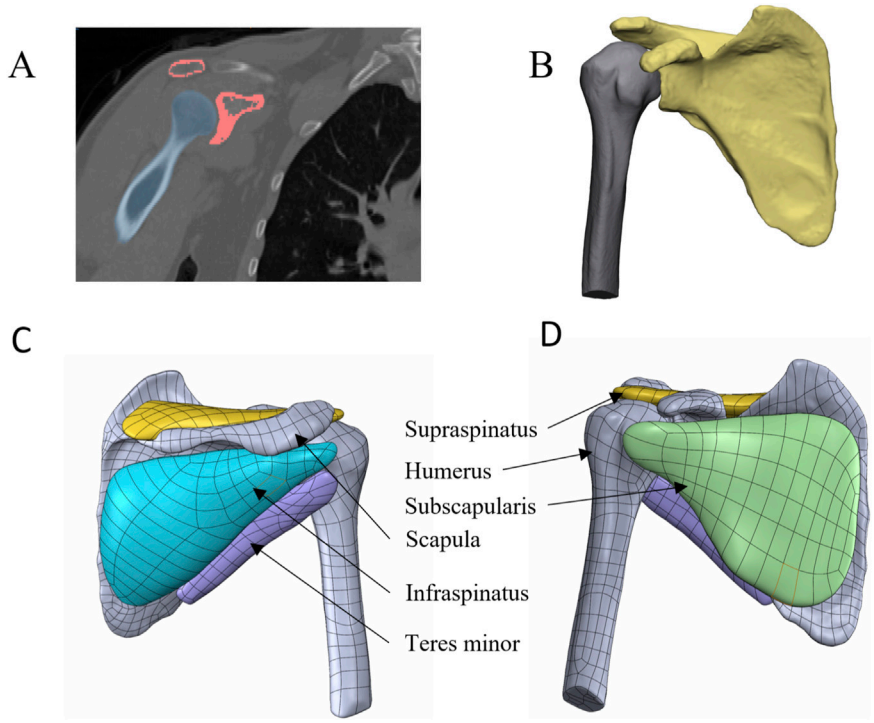


FIGURE 1
Workflow for establishing the finite element shoulder model. **(A)** CT threshold segmentation; **(B)** solid model reconstruction; **(C)** posterior view of the shoulder joint; **(D)** anterior view of the shoulder joint.

supraspinatus repair (Quental et al., 2020; Yang et al., 2023a; Büchler et al., 2002), the medial base of the scapula was fully fixed to simulate stable scapulothoracic support. Tied constraints were applied at bone-tendon attachment regions to represent firm attachment. Bone-cartilage and cartilage-cartilage interfaces were modeled using frictionless contact to reflect low-friction sliding between articular surfaces. In addition to fixation of the medial scapular border, two distributed surface loads were applied to the humerus. The distal load was applied to a predefined surface at the distal end of the humeral shaft to represent external loading transmitted through the humerus. The proximal load was applied to a predefined surface of the humeral head to provide a joint-directed compressive component. Both loads were prescribed as applied loads of 50 N acting inward along the local normal directions of the corresponding

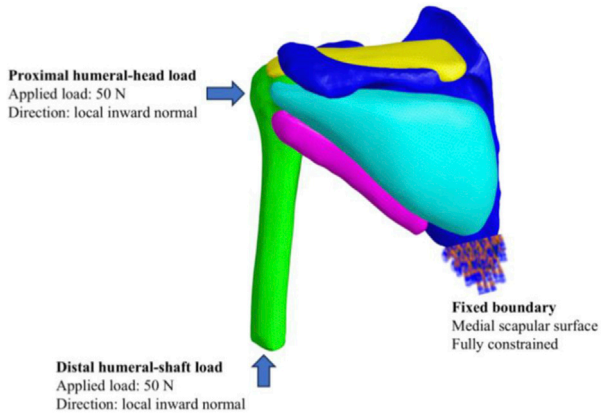


FIGURE 2
Boundary conditions and loading applied to the finite element model.

loading surfaces. Therefore, the loading orientations followed the local surface normals after the humerus was positioned at each abduction angle.

2.4 Model verification

The validity of the finite element model of suture-bridge shoulder repair was assessed from four aspects: geometric reconstruction, mesh convergence, boundary-condition definition, and mechanical response trends. According to the principles of verification and validation in computational biomechanics, the reliability of a finite element model should not be judged from a single output alone; instead, it should be evaluated comprehensively using model anatomy, mesh independence, boundary-condition sources, and agreement with previous experimental or finite element findings (Anderson et al., 2007). In the present study, the scapula, humerus, articular cartilage, and supraspinatus tendon were reconstructed from volunteer imaging data. The model components, contact definitions, and expected force-response trends of the rotator cuff were established with reference to previous shoulder finite element and cadaveric biomechanical studies (Yang et al., 2023a; Büchler et al., 2002; Turkel et al., 1981).

A controlled mesh-convergence analysis was defined using the representative 20°/60 N condition, with geometry, materials, contacts, constraints, pretension, loading, solver settings, and post-processing regions held constant. As Table 2 showed, mesh sizes of 1.5, 1.0, and 0.8 mm contained 552,396, 1,609,968, and 3,265,434 elements, respectively. When the mesh was refined from 1.0 to 0.8 mm, the relative changes in mean contact pressure, peak tendon-side von Mises stress, and mean relative interfacial displacement were 0.98%, 1.92%, and 0.75%, respectively. Because all three changes were below 2%, the 1.0-mm mesh was used for the subsequent analyses.

The boundary conditions and contact definitions in this study were also based on previous shoulder finite element models. In the shoulder model established by Büchler et al., the normal glenohumeral contact forces in a normal shoulder were approximately 31.9 N at 30° of external rotation and 31.5 N at 40° of internal rotation, indicating that finite element models can be used to evaluate glenohumeral contact force, contact pressure, and bone stress distribution (Büchler et al., 2002). In the present study, the medial base of the scapula was fixed, frictionless contact was used between bone-cartilage and cartilage-cartilage interfaces, and tied constraints were applied at bone-tendon attachment regions. These settings are consistent with commonly used approaches in previous shoulder finite element studies (Yang et al., 2023a; Büchler et al., 2002).

Regarding mechanical response trends, previous studies have shown that shoulder abduction and external loading substantially increase supraspinatus tendon stress, with high-stress regions often located near the supraspinatus insertion and tendon-bone attachment area (Yang et al., 2023a; Yang et al., 2023b). Yang et al. reported that, under 30° of abduction with loading, the maximum von Mises stress of the supraspinatus increased by approximately 75% compared with the unloaded condition. Another finite element study of shoulder abduction reported that articular-side stress of the supraspinatus was approximately 15.5 MPa at 90° of abduction and increased to 22.2 MPa after loading (Yang et al., 2023b). In the present study, tendon-bone contact pressure and tendon-side von Mises stress increased with increasing pretension, and stress concentration was mainly located around the suture passage region and tendon-bone interface. These findings are consistent with the stress-concentration sites and response trends reported in previous rotator cuff finite element studies (Yang et al., 2023a; Yang et al., 2023b; Redepenning et al., 2020). Thus, the present finite element model was considered suitable for analyzing the effects of different pretensions on tendon-bone contact pressure, tendon stress, and relative interfacial displacement.

2.5 Simulation of full-thickness supraspinatus tear and suture-bridge repair

Based on the validated intact shoulder model, SolidWorks was used to create a full-thickness supraspinatus tendon tear. The tear was located approximately 1.5 cm medial to the insertion on the greater tuberosity. According to the clinical double-row suture-bridge repair technique, two medial-row anchors and two lateral-row anchors were placed at the medial and lateral margins of the greater tuberosity footprint, respectively. Sutures passed from the medial-row anchors through the supraspinatus tendon, crossed over the tendon surface, and were fixed to the lateral-row anchors to form a double-row suture-bridge repair construct.

To evaluate the biomechanical influence of different suture pretensions, pretension was applied along the suture direction at the free suture end before fixation to the lateral-row anchor (Veeger, 2000). Seven total pretension levels were defined: 30, 40, 50, 60, 70, 80, and 90 N. The humerus was then positioned relative to the scapula to create three early postoperative abduction positions, 10°, 20°, and 30°, as shown in Figures 3A,B, thereby simulating the mechanical response of the repair construct under different shoulder abduction angles. The three abduction angles were selected to assess the positional consistency of pretension-dependent responses within a standardized low-abduction range, rather than to compare shoulder positions or determine an optimal abduction angle. At the analyzed increment, effective tendon-bone

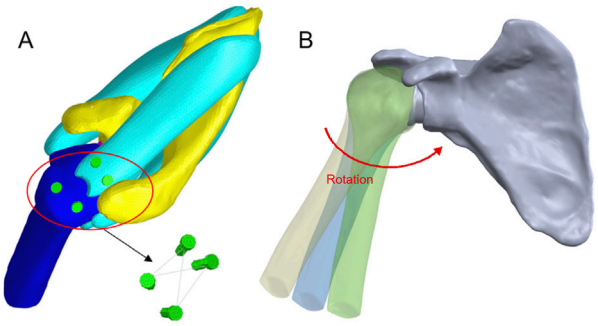


FIGURE 3
Suture-bridge repair model and abduction-angle settings. **(A)** suture-bridge repair model; **(B)** humeral rotation during shoulder abduction.

contact area was calculated by summing the areas of the selected triangular surface facets within the footprint for which CPRESS >0. Facets with CPRESS = 0 were considered noncontacting and were excluded. Mean contact pressure was calculated as the area-weighted mean CPRESS over the same active facets.

3 Results

3.1 Tendon-bone contact pressure under different pretensions and abduction angles

The tendon-bone contact-pressure distributions under different abduction angles and pretension levels are shown in Figure 4. In all conditions, contact pressure was mainly distributed over the greater tuberosity footprint and the region compressed by the suture bridge, gradually decreasing from the central contact area toward the periphery. As pretension increased from 30 N to 90 N, local peak pressure increased at all three abduction angles. The contact-pressure maps showed two spatially distinct features. A narrow band of elevated pressure was observed along the medial, glenoid-facing margin of the tendon-bone interface across the pretension levels and was interpreted as localized edge contact at the boundary of the contacting region. In contrast, the broader pretension-dependent increase in contact pressure occurred mainly around the suture-compression region. This change was most pronounced at 20° of abduction, where pressure elevation around the suture-compression region became more evident at 70–90 N.

Mean contact pressure increased continuously with pretension. At 10° of abduction, mean contact pressure increased from 0.280 MPa to 0.600 MPa; at 20°, it increased from 0.382 MPa to 0.856 MPa; and at 30°, it increased from 0.344 MPa to 0.690 MPa. The corresponding increases were 114.3%, 123.9%, and 100.8%, respectively.

Peak contact pressure also increased with pretension, although the magnitude of increase differed among abduction angles. At 10°, peak contact pressure increased from 2.281 MPa to 2.974 MPa; at 30°, it increased from 2.770 MPa to 2.988 MPa; whereas at 20°, it increased from 3.048 MPa to 9.706 MPa, corresponding to a 218.4% increase.

The effective contact area did not increase uniformly with pretension. At 10° of abduction, the effective contact area remained relatively stable, ranging from 229.0 to 232.0 mm^2. At 30°, the contact area increased from 212.5 mm^2 at 30 N to 217.2 mm^2 at 50–60 N and then decreased slightly. At 20°, the contact area decreased from 181.4 mm^2 to 169.5 mm^2 as pretension increased. These findings indicate that the increase in peak pressure at 20° was accompanied by a reduction in contact area, suggesting more evident local pressure concentration.

3.2 Tendon-side von mises stress under different pretensions and abduction angles

High tendon-side von Mises stress was mainly located around the suture holes and adjacent load-bearing bands. Under low pretension, the high-stress region was relatively limited. As pretension increased, yellow-to-red stress regions gradually expanded. Compared with the contact-pressure maps, the stress maps (Figure 5) more directly demonstrated local stress concentration around the suture passage sites.

Peak von Mises stress increased continuously with pretension. At 10° of abduction, peak stress increased from 1.581 MPa to 3.917 MPa; at 20°, it increased from 1.516 MPa to 3.475 MPa; and at 30°, it increased from 2.135 MPa to 4.533 MPa. The 30° condition consistently showed the highest peak stress among the three abduction angles.

Mean von Mises stress showed the same increasing trend. At 10°, the mean stress increased from 0.343 MPa to 0.711 MPa; at 20°, from 0.397 MPa to 0.811 MPa; and at 30°, from 0.489 MPa to 0.973 MPa. These results indicate that increasing pretension simultaneously elevated the overall tendon stress level and local peak stress.

3.3 Quantitative changes in key mechanical indicators

The mean contact pressure, effective contact area, mean relative displacement, and peak tendon-side von Mises stress under different pretensions and abduction angles are compared in Figures 6A–D. Both mean contact pressure and tendon-side stress increased with pretension under all three abduction conditions. From 30 N to 60 N, the increase in mean contact pressure was relatively pronounced, indicating progressive establishment of tendon-bone compression. From 60 N to 90 N, mean contact pressure continued to increase, but peak contact pressure, mean relative displacement, and peak tendon-side von Mises stress also increased concurrently.

Joint consideration of contact pressure, effective contact area, relative interfacial displacement, and tendon-side stress showed that 50–60 N occupied an intermediate range among the tested pretension levels. Compared with 30–40 N, this range produced greater tendon-bone compression. Compared with 70–90 N, it was associated with lower peak contact pressure, relative interfacial displacement, and tendon-side stress. Thus, within the simulated abduction conditions, 50–60 N showed a comparatively balanced mechanical response across the evaluated indicators. The effects of increasing pretension were interpreted by considering the gain in interface compression together with the accompanying changes in contact area, displacement, and tendon-side stress.

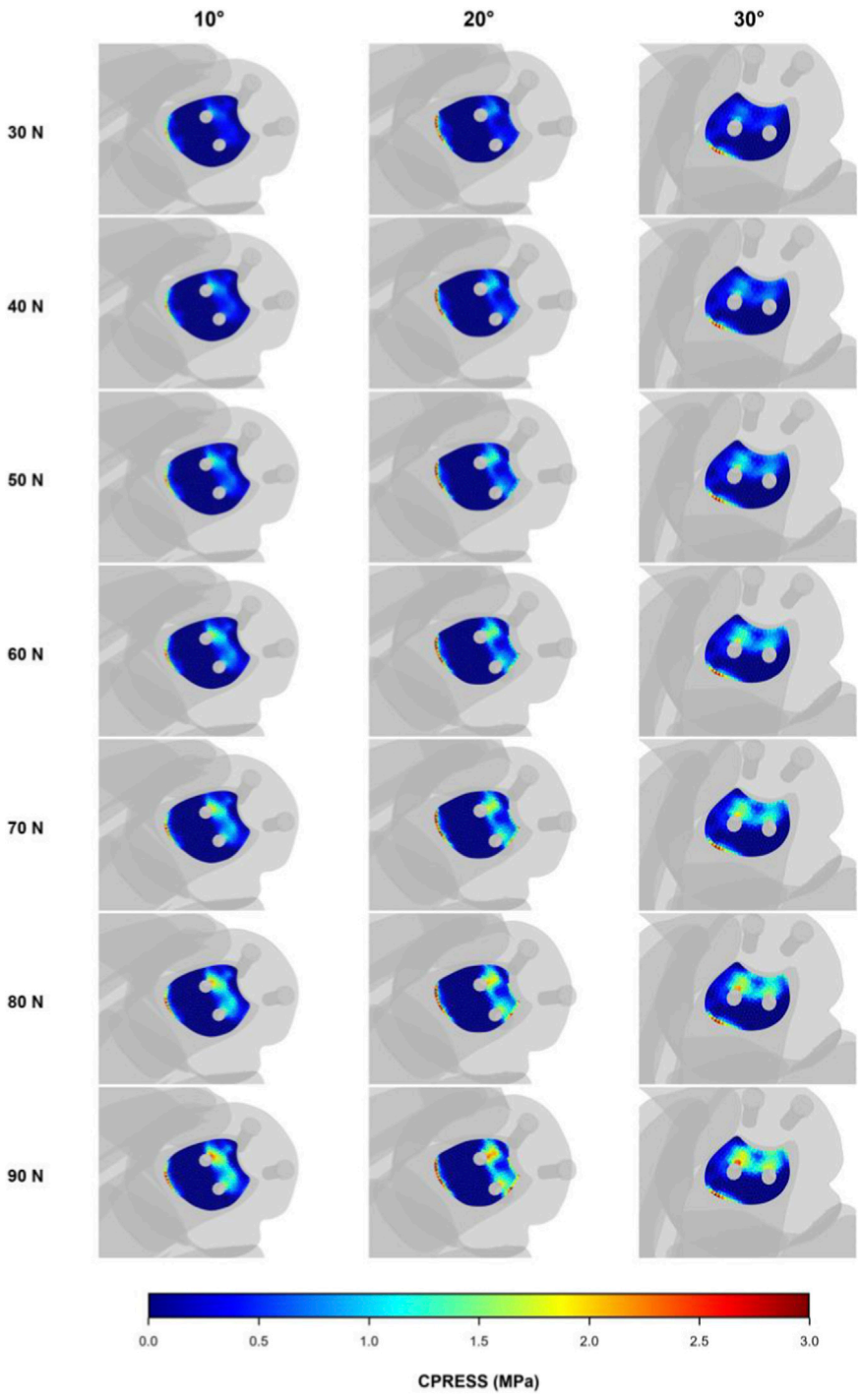


FIGURE 4
Contact-pressure distribution at the tendon-bone interface.

3.4 Integrated trade-off evaluation of pretension conditions

To avoid simply listing numerical values for all conditions, the four key indicators at the tendon-bone region were integrated into a benefit-risk distribution plot (Figure 7). In this plot, the horizontal axis represents mean contact pressure, reflecting the compressive benefit at the tendon-bone interface; the vertical axis represents peak tendon-side von Mises stress, reflecting local tendon stress risk; bubble size represents effective contact area; color represents mean relative displacement; and the number within each bubble indicates the corresponding pretension. This format allows interface compression, contact range, interfacial micromotion, and tendon stress to be presented within a single coordinate system.

As shown in Figure 7, the 30–40 N conditions were mainly located in the low mean-contact-pressure region, indicating

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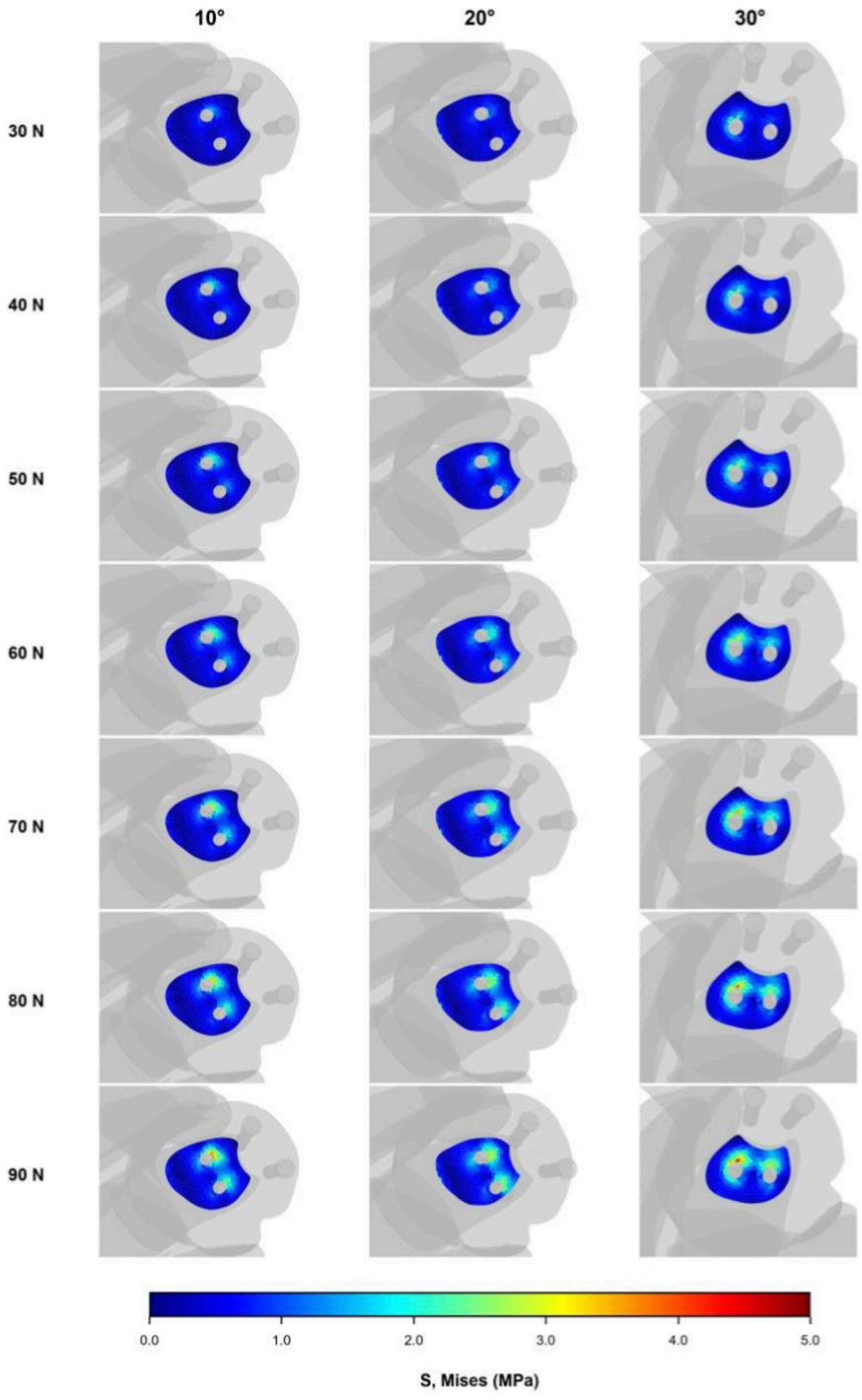


FIGURE 5
Tendon-side von Mises stress distribution of the supraspinatus tendon.

insufficient compression of the tendon-bone footprint. Mean contact pressure generally increased with increasing pretension at all three abduction angles. However, the increase within the 50–60 N range was most pronounced at 20° of abduction, whereas the corresponding changes at 10° and 30° were more gradual. This finding indicates that the pressure response to pretension was dependent on the simulated abduction angle. When pretension increased to 70–90 N, the data points shifted toward higher contact pressure and higher tendon stress, the bubble color became bluer, indicating increased relative displacement, and the bubble size decreased, indicating reduced contact area. Across the tested models, 50–60 N represented an intermediate pretension range with greater interface compression than 30–40 N, although relative displacement and tendon-side stress were also higher. However, the magnitude of these responses varied with abduction angle, and the increase in mean contact pressure

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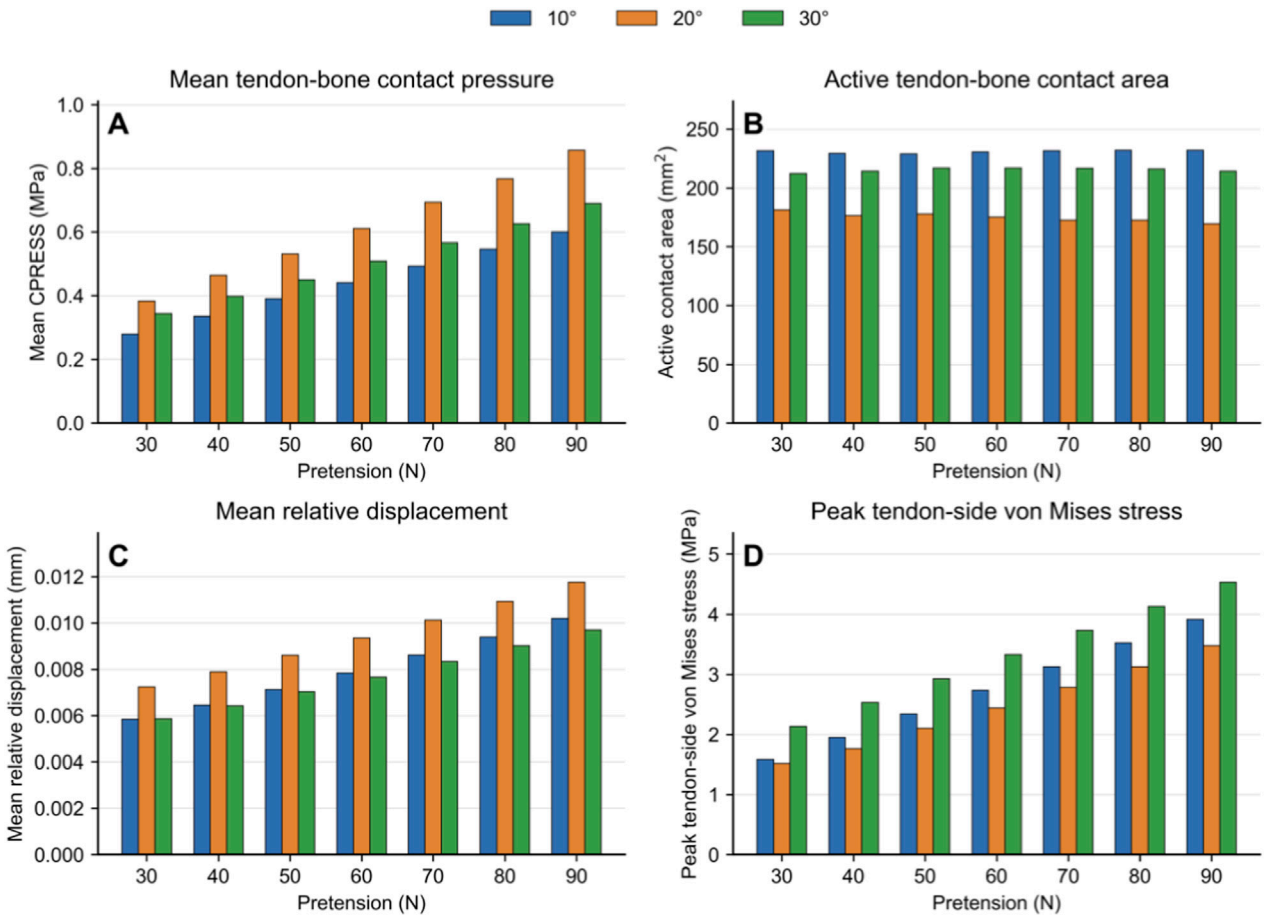


FIGURE 6
Grouped bar charts of key mechanical indicators at the tendon-bone interface. **(A)** mean contact pressure; **(B)** effective contact area; **(C)** mean relative displacement; **(D)** peak tendon-side von Mises stress.

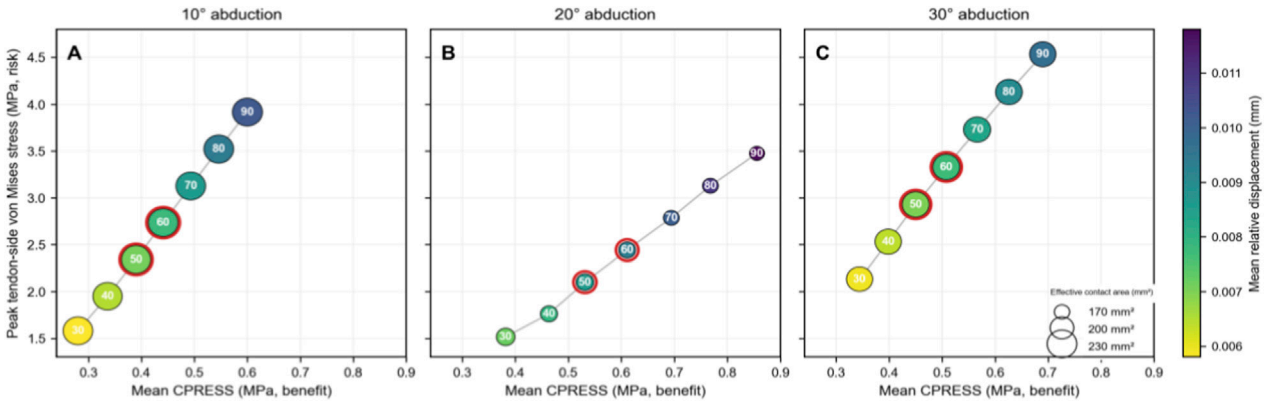


FIGURE 7
Integrated benefit-risk distribution of pretension conditions.

within this range was most pronounced at 20°. Therefore, 50–60 N should not be interpreted as an equally optimal range for all abduction angles or as a universal clinical target, but rather as a condition-dependent biomechanical reference within the simulated configurations.

4 Discussion

This study investigated suture pretension as a controllable factor in suture-bridge repair by comparing the tendon-bone mechanical responses produced by 30–90 N pretension under three shoulder abduction positions. The results showed that increasing pretension improved mean contact pressure at the tendon-bone interface, but it also increased tendon-side von Mises stress, relative interfacial displacement, and local peak pressure. Around 50–60 N, the repair construct showed a relatively balanced mechanical profile between interface compression and tissue loading. It should be emphasized that the 10°, 20°, and 30° abduction positions were included to broaden the range of early postoperative positions analyzed, rather than to determine whether one abduction angle was clinically superior to another.

From a mechanical perspective, low pretension may provide insufficient compression of the tendon stump, resulting in low mean contact pressure and potentially inadequate footprint apposition (Baums et al., 2010; Chernchujit et al., 2023). As pretension increases, suture tension is progressively converted into normal compressive force between the tendon and bone bed, thereby increasing interface pressure. However, with further pretension increases, the additional pressure gain was expressed more as elevation of local peak pressure than as expansion of the effective contact area. This suggests that excessive tension may concentrate load around the suture-compression region, suture holes, and footprint margins, thereby increasing the risk of local high pressure and tissue stress concentration.

The tendon-side von Mises stress pattern further supports this interpretation. In this study, high-stress regions were mainly located around the suture holes and adjacent load-bearing bands, and these regions expanded with increasing pretension. This distribution is consistent with the clinical observation that suture cut-through and retears near the medial row often occur at the suture-tendon contact region (Sano et al., 2017). Therefore, pretension should not be evaluated simply according to the principle that higher contact pressure is better. Interface compression, contact area, micromotion control, and local tendon stress should be assessed together.

Previous studies have examined individual components of this biomechanical problem. Park et al. (Park et al., 2015) investigated bridging-suture tension in a cadaveric model, while Andres et al. (Andres et al., 2010) evaluated the effects of tension and abduction on footprint compression in an ovine model. Finite element studies by Sano et al. (Sano et al., 2007) and Quental et al. (Quental et al., 2020) characterized tendon stress distribution or compared different repair configurations. Building on these studies, the present study evaluated seven pretension levels across three abduction positions and simultaneously quantified contact pressure, effective contact area, relative interfacial displacement, and tendon-side stress. This integrated assessment provides additional evidence for interpreting the mechanical trade-offs associated with suture pretension.

Across the tested models, 50–60 N represented an intermediate pretension range with increased interface compression relative to 30–40 N, while relative displacement and tendon-side stress were also higher. However, the magnitude of these responses varied with abduction angle, and the increase in mean contact pressure within this range was most pronounced at 20°. Therefore, 50–60 N should not be interpreted as an equally optimal range for all abduction angles or as a universal clinical target, but rather as a condition-dependent biomechanical reference within the simulated configurations.

This study has several limitations. First, the model was reconstructed from the imaging data of a single healthy volunteer and therefore did not account for differences in age, bone mineral density, rotator cuff degeneration, or individual anatomy. Second, the supraspinatus tendon was simplified as a homogeneous, isotropic, and linearly elastic material and therefore did not reproduce the fiber-dependent, strain-dependent, and time-dependent behavior of the native tendon. This simplification may affect the absolute magnitude and spatial distribution of tendon stress and deformation, particularly around the suture passage sites, and may consequently influence the calculated tendon-bone contact and relative interfacial displacement. Because the same constitutive formulation was applied across all simulated conditions, the pretension-dependent trends were interpreted comparatively; however, the absolute tendon-related values should be interpreted cautiously. Future studies should incorporate anisotropic, nonlinear, and viscoelastic tendon material models. Third, this study was limited to static loading at 10°, 20°, and 30° of abduction. Shoulder rotation, flexion, larger abduction angles, and dynamic or cyclic motion may alter tendon orientation, muscle loading, footprint contact, and local stress distribution. Therefore, the quantitative findings should not be extrapolated directly to shoulder positions or movements that were not simulated. Although the general trade-off between interface compression and local tissue loading may persist, its magnitude and the corresponding pretension level may be position dependent. The model also did not include fatigue failure, tissue creep, or material changes during tendon-bone healing. Fourth, the contact definitions, tied constraints, and loading boundary conditions were idealized and cannot fully reproduce the real surgical and rehabilitation environment. Finally, the 50–60 N pretension range was predicted by finite element analysis and requires further validation using cadaveric biomechanical experiments and clinical follow-up (Anderson et al., 2007; Redepenning et al., 2020).

In summary, the present study indicates that a mechanical balance zone exists for pretension in suture-bridge repair. Moderate pretension can improve tendon-bone compression, whereas excessive pretension may increase local high pressure and tendon stress concentration. These results may provide a useful reference for pretension control in suture-bridge repair and for subsequent experimental studies.

5 Conclusion

This finite element study compared tendon-bone mechanical responses after suture-bridge repair of supraspinatus tendon

tears under different pretension levels. Appropriate increases in pretension improved mean contact pressure and footprint compression, whereas excessive pretension increased peak pressure, relative displacement, and tendon stress and reduced effective contact area. Within the tested configurations, 50–60 N represented an intermediate, condition-dependent range rather than an equally optimal range for all abduction angles or a universal clinical target. These findings provide a quantitative biomechanical reference for intraoperative tension control, although further cadaveric, animal, and clinical validation is required.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The studies involving humans were approved by Ethics Committee of Xinjiang Uygur Autonomous Region Hospital of Traditional Chinese Medicine, Urumqi, Xinjiang, China. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

HW: Writing – original draft, Writing – review and editing. YL: Writing – review and editing. LM: Writing – review and editing. CG: Writing – original draft. YH: Writing – original draft, Writing – review and editing. RS: Writing – review and editing, Writing – original draft.

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Conflict of interest

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